



CHAPTER 1- CONTINUED

DATABASE MODELS

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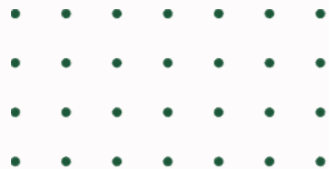
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01

Data Model

KDK College of Engineering Nagpur.

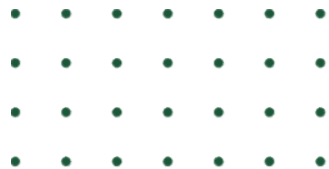
<https://kdkce.edu.in/writereaddata/fckimagefile/U1-Data%20Model.pdf>

02

DATABASE SYSTEM CONCEPTS.

Abraham Silberschatz.

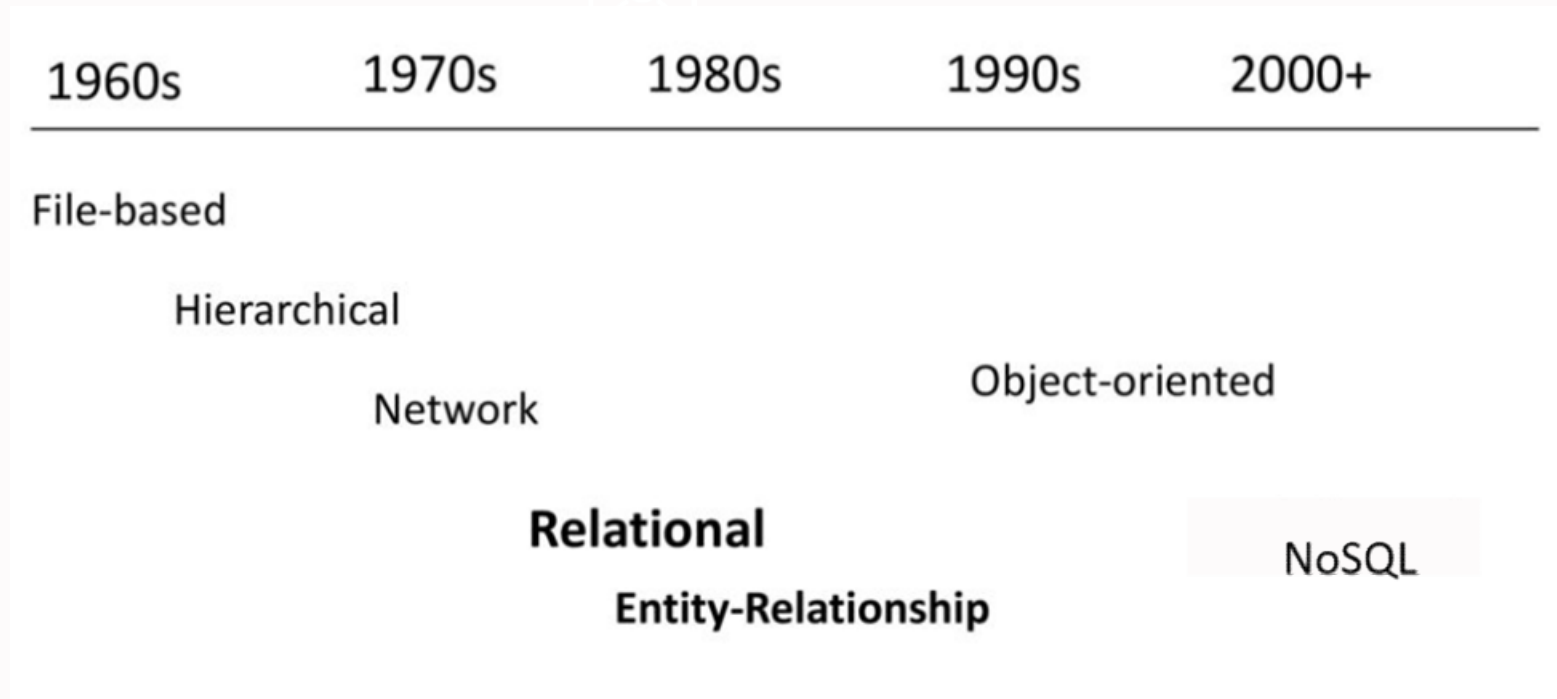
<https://people.vts.su.ac.rs/~peti/Baze%20odataka/Literatura/Silberschatz-Database%20System%20Concepts%206th%20ed.pdf>



What is a database Model?

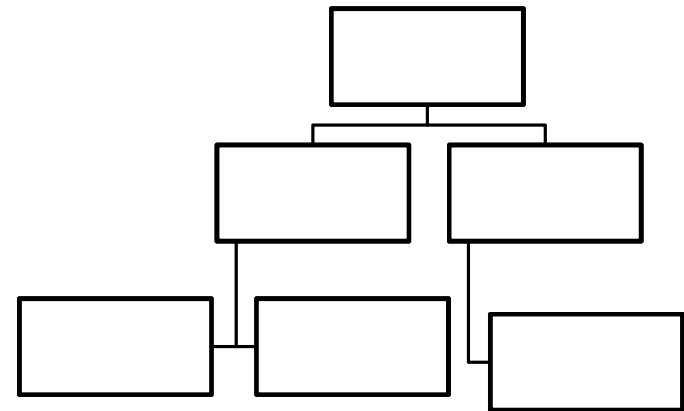
- A model is a **representation of reality**, 'real world' objects and events, and their associations.
- A data model defines the **data elements** and the **relationships** between the data elements.
- Data Models show how data is stored, organized, related, accessed and updated in the database management system.

Evolution of database models



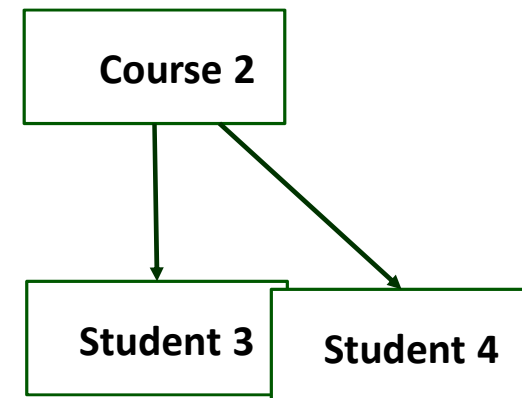
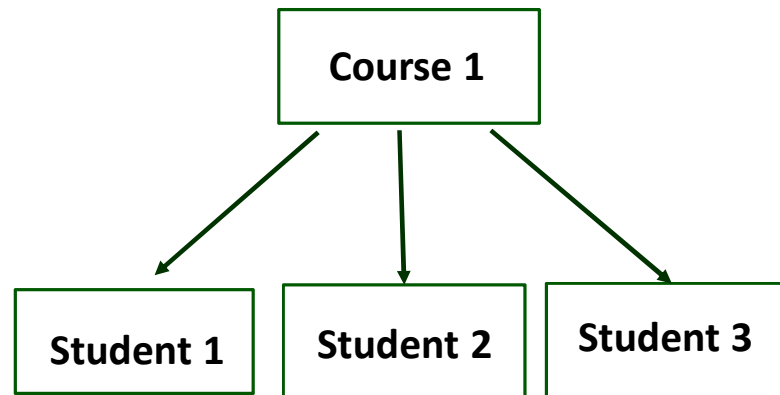
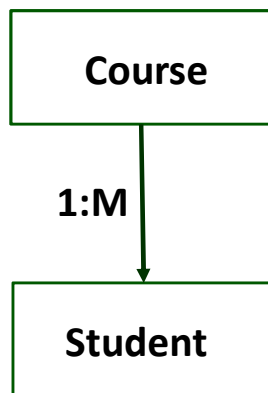
Hierarchical model

- Organizes the data in the hierarchical **tree** structure.
- Assume data relationships are hierarchical
 - **One-to-Many (1:M) relationships**
 - Each parent has many children
 - Each child has only one parent
- **Pointers** are used to link the parent node with the child node (**records**) and are used to navigate between the stored data.
- One of the first hierarchical databases: Information Management System (IMS)



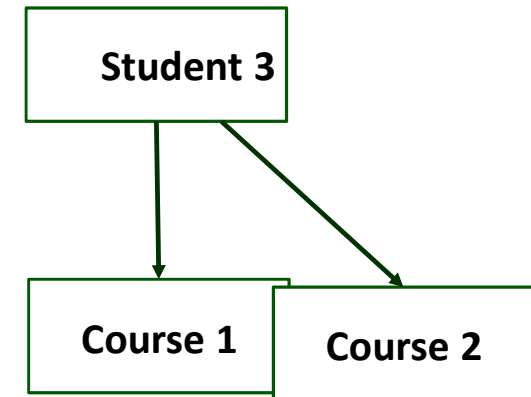
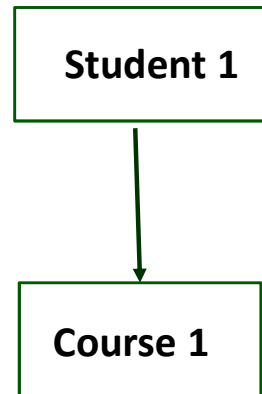
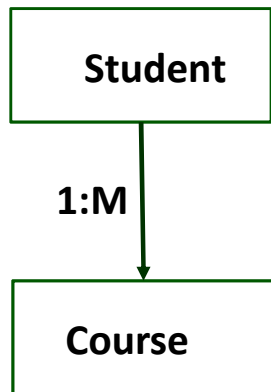
Hierarchical model – Example 1

Students subscribed in a course



Hierarchical model – Example 2

Courses of each students



Hierarchical model – Pros & Cons

● Advantages

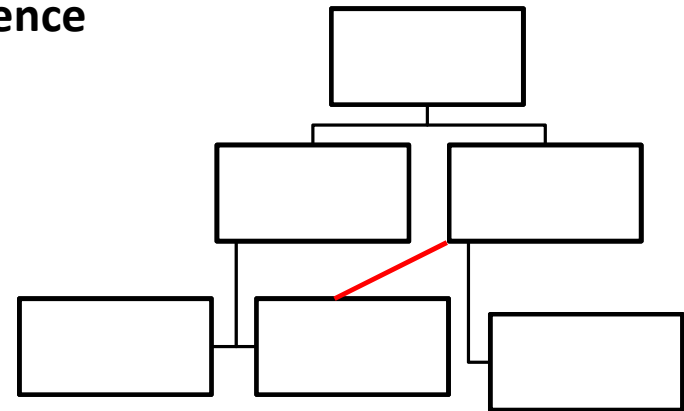
- Very simple and fast to traverse.

● Disadvantage

- Complex relationships are not supported (no child with more than one parent).
- Data redundancy => multiple updates
- Can't insert data of child without parent (in ex 1: can't insert student without course)
- If a parent node is deleted then the child node is automatically deleted.

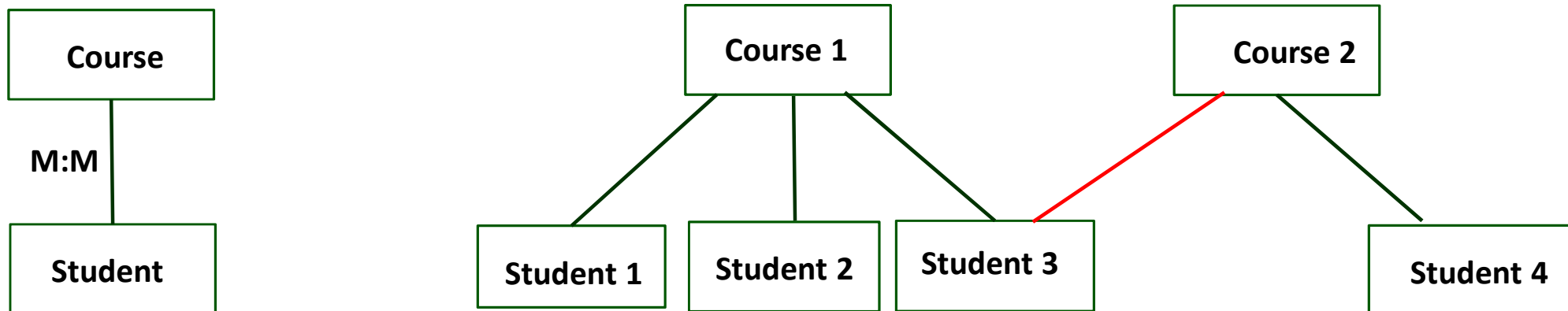
Network model

- It is an extension of the hierarchical model. The difference is that an element can have **more than one parent**.
- A network structure allows: one-to-one (1:1), one-to-many (1:M), many-to-one (M:1) and **many-to-many (M:M) relationships**.
- **Circular Linked List** are used to link members (records) of the same owner and is used to navigate between the stored data.
- One of the first network databases: Integrated Data Store (IDS)



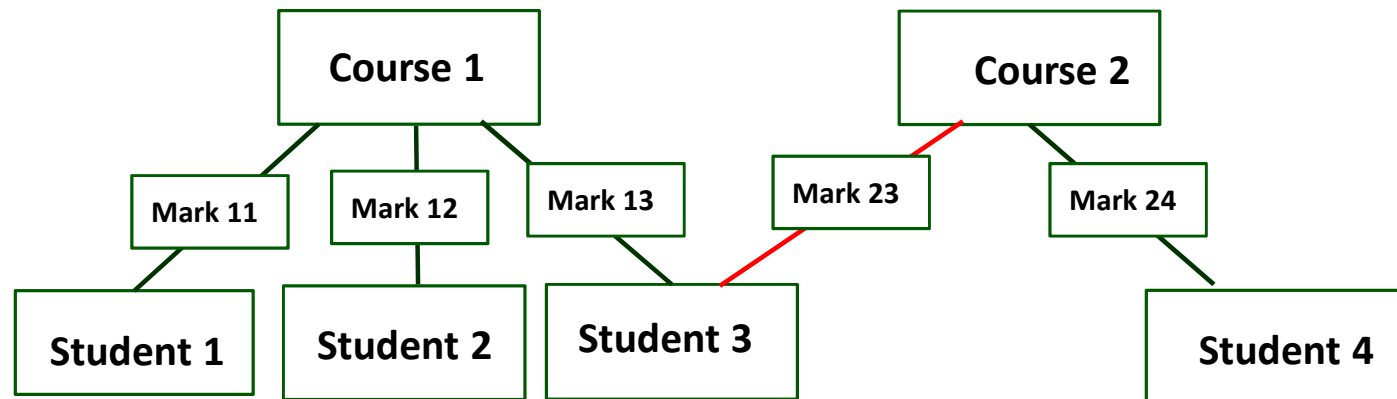
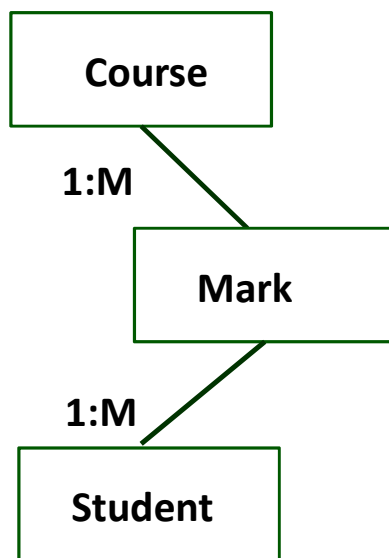
Network model – Example 1

Students subscribed in a course and Courses of a student



Network model – Example 1

Represent Evaluations which depend on Course and Student: A mark for each student in each course



Network model – Pros & Cons

● Advantages

- Represent more relationships.
- Fast data access: more paths
- No redundancy

● Disadvantage

- Complex to handle all relationships.

Relational model

- Relational model stores data in the form of **two-dimensional tables**.
- The tables are also called **relations**.
- The **columns define the structure** of an element and the **rows defines the values** of an element.
- A table can be related to another by using common fields.
- Example of relational databases: Oracle, MySQL, Postgres.

Employees			
Emp_ID	Emp_Name	Salary	Department
4182	A. Singh	36,003.12	Development
5542	M. Lee	39,818.15	Development
3351	T. Esterhazy	31,919.18	Development
6164	A. Barraclough	38,002.05	Marketing
9095	N. Prabakar	36,003.12	Research

Departments	
Department	Budget
Development	500,000
Marketing	150,000
Research	100,000

Relational model – Example 1

Students related to a Department

Student table

Stu_ID	S_Name	S_Address	Birthdate	Sex	Dept_ID
121	Anita	41, Kendy Av.	15-07-1981	F	1
122	Dravid	22, Basant Av.	23-06-1980	M	2
125	Shallu	120, St.Thomas Av.	19-09-1982	F	2
131	Atul	58, Lwr.Rd.	27-08-1978	M	1

Department table

Dept_ID	Dept_Name	Acronym
1	Intelligence artificielle et Science de Données	IA&SD
2	Système Informatique	SIQ

Relational model – Example 2

Students courses

Student table

Stu_ID	S_Name	S_Address	Birthdate	Sex	Dept_ID
121	Anita	41, Kendy Av.	15-07-1981	F	1
122	Dravid	22, Basant Av.	23-06-1980	M	2
125	Shallu	120, St.Thomas Av.	19-09-1982	F	2
131	Atul	58, Lwr.Rd.	27-08-1978	M	1

Courses table

Course_ID	Course_Name	Acronym	Coef
1	Algorithmic and complexity	ALGO	5
2	Operating System	OS	3
3	English	EN	1

Student_Courses table

Stu_ID	Course_ID	Mark
121	1	15
121	3	12
122	3	10
125	2	17
125	1	18

Relational model – Pros & Cons

● Advantages

- **Simplicity & Ease of Use** : The tabular structure is intuitive and easy to understand compared to hierarchical or network models.
- **Scalable**: we can easily add columns and rows.
- **Strong mathematic background**: we know how to manipulate relations.

● Disadvantage

- **See later.**

Entity Relationship model

- Entity-Relationship Model (ER) represent the real-world problem in graphical form to make it easy to understand.

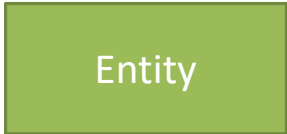
ER is a tool for conceptual data modeling (high level representation)

- ER Diagram

Entities: Entity is a real-world thing. It can be a person, place, or even a concept.

Attributes: An entity contains a real-world property called attribute. This is the characteristics of that attribute.

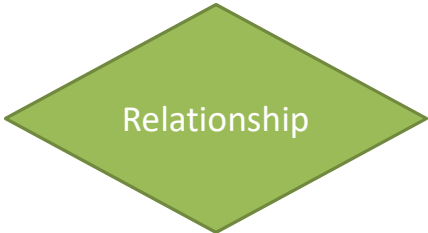
Relationship: Relationship tells how two attributes are related. Example: Teacher works for a department.



Entity

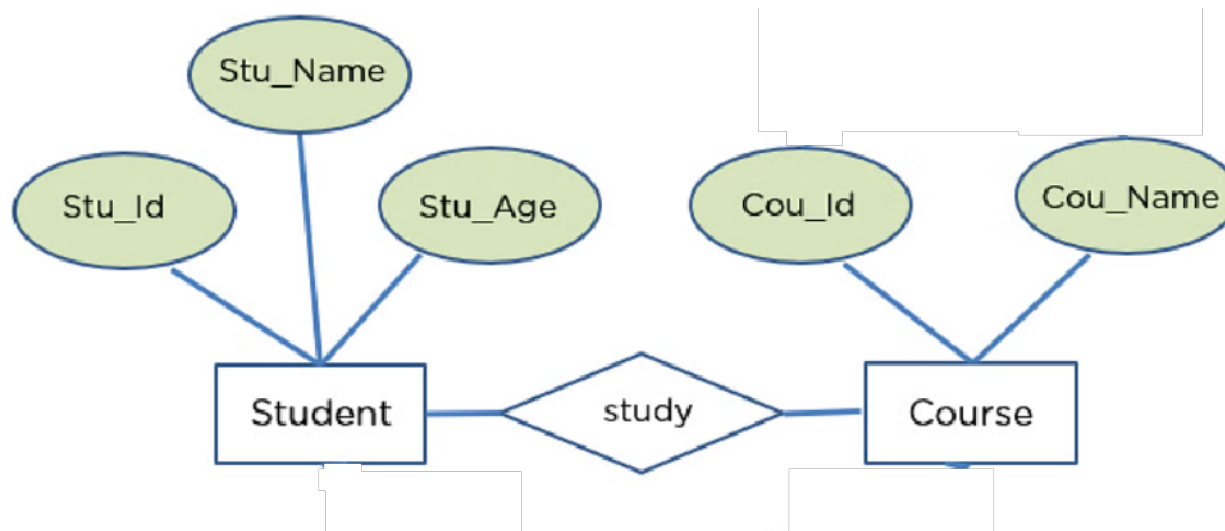


Attribute

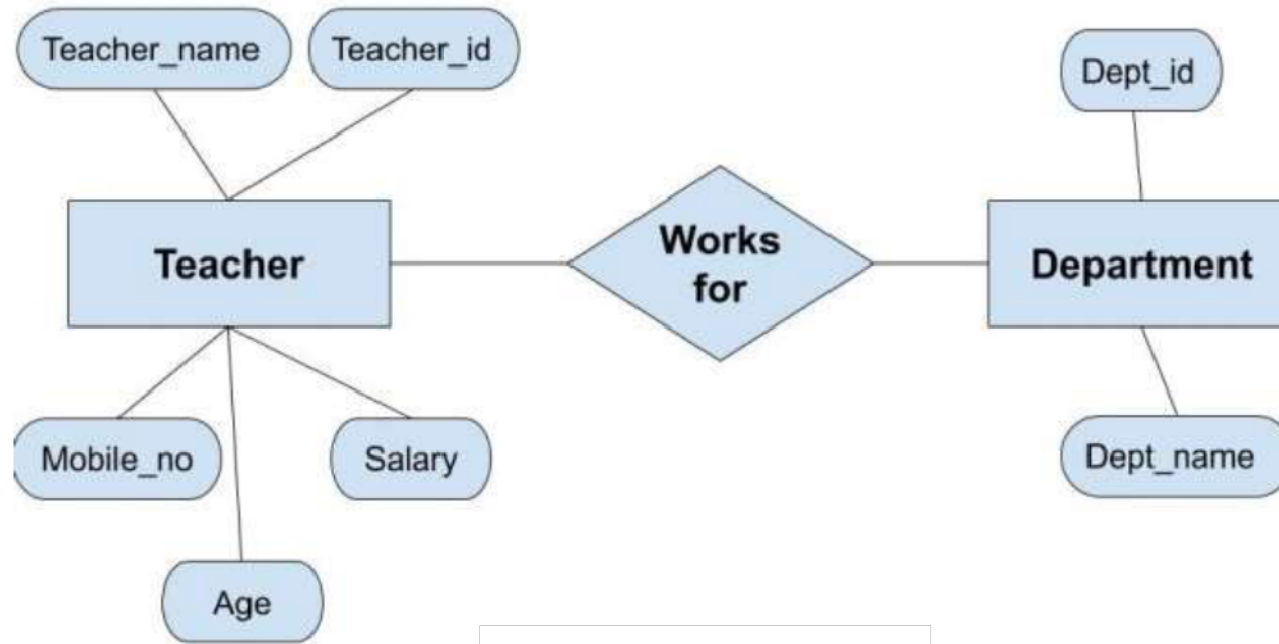


Relationship

Entity Relationship model - Example



Entity Relationship model - Example



Entity Relationship model – Cardinality

Cardinality in the ER model is the **number of instances** of one entity that can be associated with the instances of another entity in a relationship.

Many-to-many (M:N) : A student study multiple courses. A course is attended by multiple students



One-to-many (1:M) : A teacher work for one department. A department have multiple teacher.



One-to-one (1:1): A director manages one university. A university is managed by one director.



Entity Relationship– Pros & Cons

● Advantages

- Extreme conceptual simplicity : easily viewed and understood.
- Integration with relational database model : enables better design via conceptual modeling.

● Disadvantage

- Incomplete model on its own: no related storage model, no data manipulation language.
It's just a representation model.

Exercise

Scenario: Hospital Management System

A **hospital** manages information about **departments, doctors, patients, and treatments**. Each department has multiple doctors, and each doctor treats multiple patients. Patients can receive multiple treatments, and each treatment is assigned to a specific doctor.

- Draw a **hierarchical schema** where:
The **root** is the hospital. Each hospital has **multiple departments**. Each department has **multiple doctors**. Each doctor has **multiple patients**.
- Design a **network schema** that represents: A **many-to-many** relationship between doctors and patients A **many-to-many** relationship between patients and treatments
- **Relational Model**: Define the **tables** for the hospital system.
- Draw an **ER diagram** representing:
Entities: **Department, Doctor, Patient, Treatment**
Relationships: **A department has doctors, a doctor treats patients, a patient receives treatments.**